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Figure 2.1

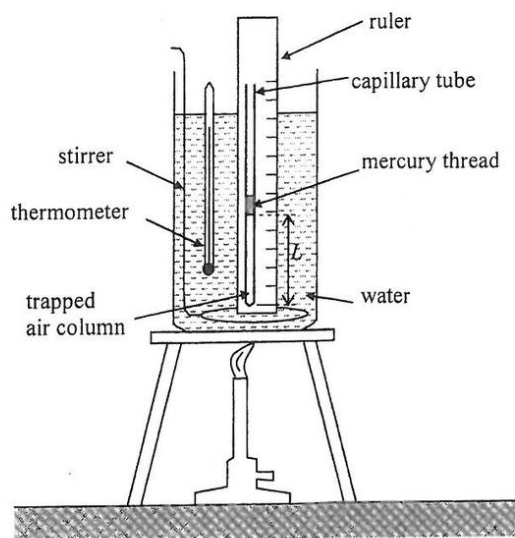


Figure 2.1 shows an air column trapped by a small mercury thread inside a uniform capillary tube. The set-up is heated by a water bath. The length of the air column L is measured at various temperature θ . Some of the results are tabulated below:

Temperature $\theta / ^\circ\text{C}$	20		92
Length of air column L / mm	64		80

- (a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped air reaches the same temperature as the water. (2 marks)

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(b) Assume that length L increases linearly with temperature θ throughout.

(i) Estimate the length of the air column when the temperature indicated by the thermometer is 65°C . (2 marks)

Answers written in the margins will not be marked.

(ii) Find the 'absolute zero' as obtained from this experiment.

(2 marks)

Answers written in the margins will not be marked.

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